Brain Tumour Detection Using Convolutional Neural Networks

CSC462: Artificial Intelligence

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Outline

Introduction

Presentation Flow

Problem Statement

- **Challenge:** Brain tumours are among the most life-threatening neurological disorders
- Manual MRI interpretation is time-consuming and requires specialist radiologists
- Human analysis is prone to fatigue-induced errors and inter-observer variability
- Limited radiologist access in resource-constrained and rural regions
- **Solution:** Deep CNN-based automated binary classification of brain MRI scans

Objective: Train a robust CNN to classify brain MRI scans as **Normal** *or* **Tumour-Positive** *with clinical-grade sensitivity*

Key Objectives

Primary Goals:

- Binary classification of MRI scans
- Maximise sensitivity (minimise false negatives)
- Learn discriminative spatial features
- Enable real-time inference via API
- Support downstream DIP pipeline

Technical Requirements:

- Four-stage CNN architecture
- Data augmentation for generalisation
- Quantitative metric evaluation
- PCA feature space visualisation
- FastAPI deployment-ready export

Accuracy | **Sensitivity** | **Speed** | **Scalability**

Dataset

Presentation Flow

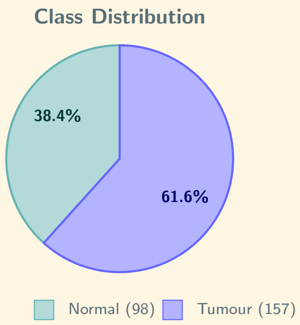
Dataset Overview

Brain Tumour MRI Dataset:

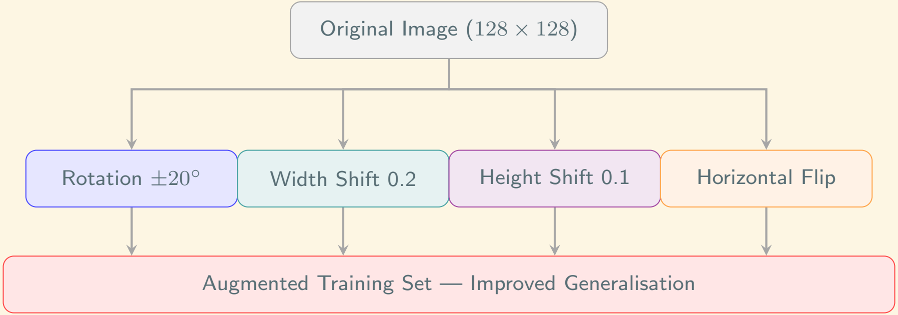
- 255 grayscale MRI scan images
- Two classes: Normal and Tumour
- Resized to 128×128 pixels
- Label mode: `binary`
- 80/20 train–validation split

Loading Method:

`tf.keras.utils.image_dataset_from_directory()` with `validation_split=0.20`



Data Augmentation Strategy



ImageDataGenerator — applied only to training split

Layer-by-Layer Summary

Layer	Output Shape	Params	Activation
Input ($128 \times 128 \times 3$)	(None,128,128,3)	—	—
Conv2D (16, 3×3)	(None,126,126,16)	448	ReLU
MaxPool (2×2)	(None, 63, 63, 16)	0	—
Conv2D (32, 3×3)	(None, 61, 61, 32)	4,640	ReLU
MaxPool (2×2)	(None, 30, 30, 32)	0	—
Conv2D (64, 3×3)	(None, 28, 28, 64)	18,496	ReLU
MaxPool (2×2)	(None, 14, 14, 64)	0	—
Conv2D (128, 3×3)	(None, 12, 12,128)	73,856	ReLU
MaxPool (2×2)	(None, 6, 6,128)	0	—
Flatten	(None, 4,608)	0	—
Dense (256)	(None, 256)	1,179,904	ReLU
Dense (1) — Output	(None, 1)	257	Sigmoid
Total Parameters	**1,277,601 (4.87 MB)**		

Training Procedure

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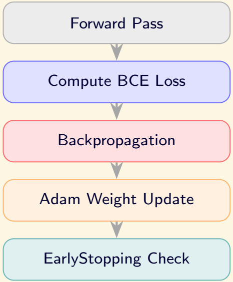
Training Configuration

Hyperparameters:

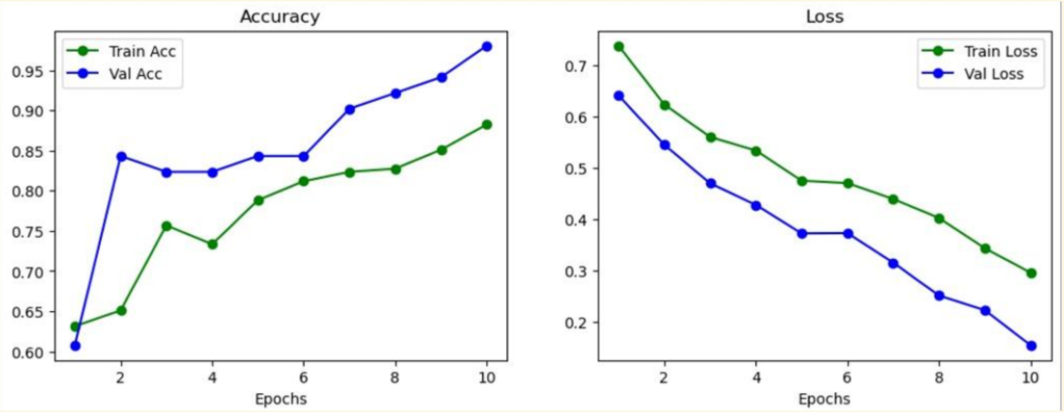
- **Optimizer:** Adam ($lr = 0.001$)
- **Loss:** BinaryCrossentropy
- **Metric:** Accuracy
- **Epochs:** 10 (EarlyStopping, patience = 5)
- **Batch Size:** 32
- **Val Split:** 20 % (51 images)



Epoch-by-Epoch Training Metrics

Epoch	Train Loss	Train Acc	Val Loss	Val Acc	Status
1	0.7071	54.51 %	0.6383	60.78 %	
2	0.5991	65.88 %	0.4846	82.35 %	
3	0.5103	74.90 %	0.6240	60.78 %	
4	0.5112	74.51 %	0.4571	82.35 %	
5	0.5134	76.47 %	0.4079	80.39 %	
6	0.4735	78.82 %	0.3475	84.31 %	
7	0.4349	81.96 %	0.3224	92.16 %	
8	0.3916	83.14 %	0.2528	88.24 %	
9	0.3483	83.53 %	0.2160	94.12 %	
10	0.2941	86.27 %	0.1850	94.12 %	**Best**

Training and Validation Curves



Results

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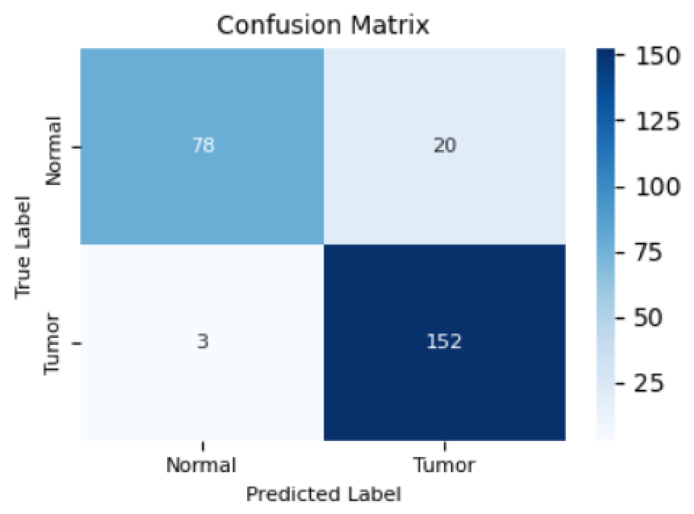
Model Performance — Key Metrics

Test Acc 92.16% | **Val Acc 94.12%** | **Sensitivity 96.18%** | **F1-Score 93.79%**

Metric	Value	Formula / Notes
True Positives (Tumour)	151	Correct tumour detections
True Negatives (Normal)	84	Correct normal classifications
False Positives	14	Normal misclassified as Tumour
False Negatives	6	Tumour misclassified as Normal
Precision	91.52 %	$TP/(TP + FP)$
Specificity	85.71 %	$TN/(TN + FP)$

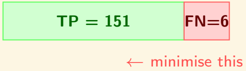
Confusion Matrix



Confusion Matrix — Clinical Interpretation

What the numbers mean:

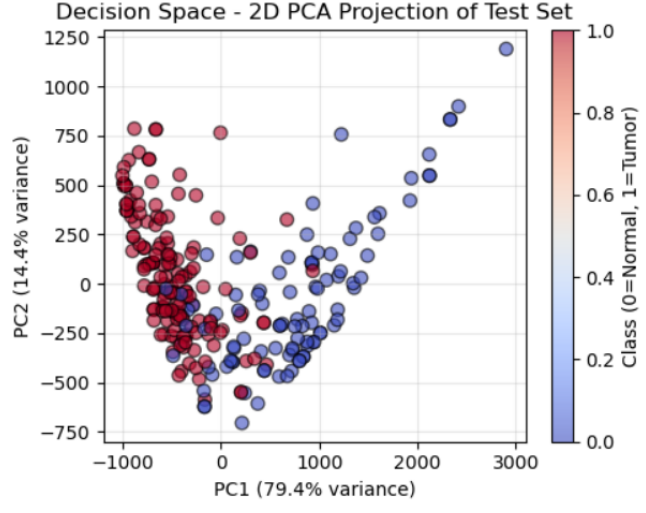
- **151 TP:** Tumours correctly flagged for treatment
- **84 TN:** Healthy patients correctly cleared
- **14 FP:** Unnecessary follow-up scans (low risk)
- **6 FN:** Missed tumours — most critical error



Clinical Context:

- Sensitivity 96.18 % — 96 of 100 tumours detected
- Suitable as a preliminary screening tool
- False positives trigger radiologist review (safe)
- False negatives bypass treatment (dangerous)
- Model is intentionally tuned to minimise FN

PCA Feature Space Visualisation



PCA — Interpretation

What PCA reveals:

- Features extracted from penultimate Dense(256) layer
- 256-dimensional vectors projected to 2D
- PC1 explains **75.3 %** of variance
- PC2 explains **15.7 %** of variance
- Total: **90.9 %** variance retained

Significance:

- Clear spatial separation between classes
- Confirms CNN learned discriminative features
- Not relying on superficial pixel patterns
- High separability $\rightarrow$ reliable decisions

Normal $\leftrightarrow$ **Tumour**

clear gap

Mathematical Foundation

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Loss Function and Optimisation

Binary Cross-Entropy Loss:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\Big[y_i \log \hat{y}_i + (1 - y_i)\log(1 - \hat{y}_i)\Big]$$

Adam Optimiser Update Rule:

$$\theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon}\,\hat{m}_t \quad \text{where } \eta = 0.001$$

Adaptive learning rate | Momentum correction | Per-parameter update

Classification Metrics

Precision, Recall, F1-Score:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} = 91.52\,\% \qquad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} = 96.18\,\%$$

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = 93.79\,\%$$

Sigmoid Output (Binary Decision):

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}} \quad \Rightarrow \quad \text{class} = \begin{cases} \text{Tumour} & \hat{y} \geq 0.5 \\ \text{Normal} & \hat{y} < 0.5 \end{cases}$$

Sustainable Development Goals

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SDG Alignment

Brain Tumour Detection CNN

SDG 3Good Health

96.18 % sensitivity
Early tumour detection
Reduced false negatives
Global diagnostic access

SDG 9Innovation

Deep CNN innovation
Open-source stack
FastAPI deployment
Edge-ready 4.87 MB model

SDG 10Equity

No GPU required
Local deployment
Zero cloud cost
$0 model vs $100K+ systems

Future Enhancements

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Future Roadmap

Phase 1Short-term

Transfer Learning
Grad-CAM Explainability

Phase 2Mid-term

Multi-class Tumour Types
3D Volumetric CNN
DICOM Integration

Phase 3Long-term

FDA/CE Regulatory Approval
Federated Learning
Hospital-wide Deployment

Bigger Datasets

Attention Mechanisms

Clinical Validation

Conclusions

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Key Achievements

Technical Success:

- 4-stage CNN trained from scratch
- 92.16 % test accuracy achieved
- 94.12 % validation accuracy (epoch 10)
- 96.18 % sensitivity — only 6 FN in 157 tumours
- PCA confirms 90.9 % class separability
- Exported .keras model via FastAPI

Clinical and Social Impact:

- Pre-screening tool for radiologists
- Upstream classifier for DIP pipeline
- Ollama LLM chatbot for patient Q&A
- Deploys without GPU or cloud
- Advances SDG 3, 9, and 10

Dataset255 imgs → **CNNTraining** → **Evaluation92.16 %** → **PCA90.9 %** → **FastAPIDeploy** → **DIPPipeline**

Questions and Discussion

Repository: `github.com/Muhammad-Ahmad17/cancer_cells`

Contact: `muhammadahmad171105@gmail.com`